

Do LLMs Rate Through the Cycle?

Procyclicality and anchoring in large language model credit ratings

Samir Chincholikar¹ · Robin Chawla^{2*}

¹ Independent researcher. ² Independent researcher.

* Corresponding author: robin.chawla.cse14@iitbhu.ac.in

Abstract

Large language models (LLMs) are increasingly proposed as credit-rating and default-assessment tools, yet whether they rate “through the cycle” or amplify it—a property prudential regulators care about but that has been asserted rather than measured—has remained untested. We hold synthetic firm fundamentals byte-identical across an ordered five-point macroeconomic-severity axis (plus a matched credit-irrelevant placebo axis) and elicit letter ratings and one-year default probabilities from five frontier models, so that any rating movement is procyclicality by construction, identified without the unobserved-fundamentals confound that limits the classical Amato–Furfine test. Across 32,000 ratings, every model is significantly procyclical (pooled 0.429 rating notches of downgrade per severity step, 95% CI [0.41, 0.45]; a 1.76-notch boom-to-recession swing), the effect is sharply asymmetric (3.2× steeper into recessions than out of them), an order of magnitude larger than the placebo, and robust to model-version fixed effects and a token-logprobability estimator. An explicit “through-the-cycle” instruction lowers the rating slope but does not restore through-the-cycle behaviour—it anchors the rating while suppressing the point-in-time default probability that should move, whereas a naive “be stable” instruction preserves that distinction better. Because model choice and prompt framing are first-order determinants of the effect, LLM-based credit assessment can silently import the capital-procyclicality channel that regulation seeks to dampen.

Keywords: large language models; credit ratings; procyclicality; through-the-cycle; prompt framing; financial regulation

JEL: G24; G28; C63

1. Introduction

Credit ratings shape regulatory capital, investment mandates, and the price of debt. A long-standing prudential concern is that ratings are procyclical—that they are downgraded in downturns and upgraded in booms even when a firm’s fundamentals have not changed—because such movements amplify the credit cycle and, through internal-ratings-based (IRB) capital rules, tighten lending precisely when the economy is weakest. Rating agencies aim to mitigate this by rating “through the cycle” (TTC): looking through transitory macroeconomic conditions to a firm’s creditworthiness over a full cycle.

Large language models are now being deployed for credit assessment, and supervisors have flagged that generative models could re-introduce procyclicality (e.g., Bank for International Settlements, 2024). The concern, however, has been asserted rather than measured: it is not known whether LLM ratings move with the macroeconomy on fixed fundamentals, by how much, or whether the standard mitigation—instructing the model to rate through the cycle—

works. Measuring this for human agencies is hard because fundamentals and the cycle move together, so the classical ordered-probit test of Amato and Furfine (2004) must proxy for unobserved fundamentals.

We provide the first controlled measurement of rating procyclicality in LLMs. Our design removes the identification problem by construction: a synthetic firm’s fundamentals are held byte-identical across an ordered macroeconomic-severity axis, so any change in the model’s rating is procyclicality, not a response to changed fundamentals. We find that all five frontier models tested are significantly procyclical, that the effect is strongly asymmetric (much steeper into recessions), and—using a placebo instruction and the models’ default-probability outputs—that an explicit through-the-cycle instruction does not restore through-the-cycle behaviour but instead induces generic anchoring.

2. Related literature

Our study connects three strands. The first is the literature on rating procyclicality and through-the-cycle methodology: Amato and Furfine (2004) formalise the ordered-probit test, Löffler (2004, 2013) characterise and question through-the-cycle rating, and the prudential stakes are set by the interaction of ratings with Basel capital rules (Kashyap and Stein, 2004; Gordy and Howells, 2006; Cantor and Packer, 1996). Our experimental design sidesteps these studies’ central difficulty—fundamentals and the cycle are observationally entangled—by holding fundamentals fixed. The second strand is the rapid adoption of LLMs in finance: language models are used for return prediction (Lopez-Lira and Tang, 2023), financial-statement analysis (Kim et al., 2024), and as simulated economic agents (Horton, 2023), and supervisors have flagged their potential to re-introduce procyclicality (Bank for International Settlements, 2024). The third strand is the behavioural limitations of LLMs: models reproduce human framing and anchoring effects (Tversky and Kahneman, 1974, 1981; Binz and Schulz, 2023), are imperfectly calibrated (Kadavath et al., 2022), and through shared foundation models can homogenise decisions (Bommasani et al., 2022). Procyclicality on fixed fundamentals is exactly such a context-driven bias, made consequential by its regulatory setting.

3. Design and identification

The unit of analysis is a rating cell = (model, prompt framing, seed, firm, macro state). We generate 80 synthetic firms with fixed fundamentals (revenue growth, EBITDA margin, leverage, interest coverage, free cash flow, liquidity, and Altman-style ratios) stratified across five credit-quality tiers and eight sectors, so the fundamentals-implied ratings span the scale (AAA–CCC). Each firm’s fundamentals are serialised once and reused byte-identically under every macro state; only a prepended macro-environment block changes. The macro axis has five ordered levels—boom, expansion, neutral, slowdown, severe recession (severity $-2 \dots +2$)—each carried by matched, number-based indicators (GDP growth, unemployment, credit spreads, default rate, lending standards) with no calendar dates, so a model cannot pattern-match a real historical episode.

Two design features sharpen identification. First, a matched-format but credit-irrelevant placebo axis (regional weather and traffic indicators, same structure and number density) measures pure frame-compliance; the net (macro minus placebo) coefficient is the demand-effect-adjusted

cyclical. Second, the per-cell random seed is keyed on the firm and framing but omits the macro state, so a firm receives an identical seed across all states—sampling noise cannot masquerade as a cycle effect. Each model returns a letter rating (mapped to an S&P notch, 1–21) and a one-year probability of default (PD).

We study four prompt framings—terse, point-in-time, explicit through-the-cycle, and a “be stable” instruction with no cycle reasoning—on five models (GPT-4o-mini, GPT-4.1-mini, GPT-4o, Gemini 3.5 Flash, Gemini 2.5 Flash-Lite), two seeds, at temperature zero. The primary estimand is the mean within-firm slope of rating notch on macro severity (notches of downgrade per +1 step; positive = procyclical), with firm-clustered (tier-stratified) bootstrap 95% confidence intervals (2{,}000 draws). We also report the boom-to-recession swing, downside vs upside half-slopes, the Amato–Furfine ordered-probit coefficient, and—for the OpenAI models—a continuous expected-notch slope computed from the rating-token log-probabilities. The full run comprises 32,000 ratings for US\$13.69, with zero parse errors; every capture is hashed and analysis is a deterministic function of the frozen data (see the Data availability statement).

4. Results

4.1. LLMs are procyclical on identical fundamentals

On byte-identical fundamentals, every model downgrades as the described macroeconomy worsens (Table 1, Figure 1a). Pooling across models under the terse framing, ratings fall by 0.429 notches per severity step (95% CI [0.41, 0.45])—a 1.76-notch swing from boom to severe recession, enough to move a BBB firm toward BB+/BB with nothing real changing. The per-model coefficient ranges from 0.230 (Gemini 3.5 Flash) to 0.775 (GPT-4o-mini), and every model’s CI excludes zero. Model identity is thus a first-order determinant of procyclicality—a roughly threefold spread—and, within OpenAI, the flagship GPT-4o (0.387) is about half as procyclical as GPT-4o-mini.

The effect is not an artefact of framing suggestibility: the matched credit-irrelevant placebo axis moves ratings only 0.042 notches per step, so the net coefficient is 0.387 (95% CI [0.36, 0.41]), still excluding zero. It is not infrastructure drift: adding served-model (system-fingerprint) fixed effects shifts the coefficient by less than 0.003. And it is not a censoring or parsing artefact: every model parses at 1.000 with the largest-cyclicity model (GPT-4o-mini) among the cleanest. For the OpenAI models, a continuous expected-notch slope computed from rating-token log-probabilities reproduces the sampled estimate (e.g., GPT-4o-mini 0.775 vs 0.765), and the Amato–Furfine ordered-probit coefficient agrees in sign and magnitude throughout.

[Insert Table 1 here] [Insert Figure 1 here]

4.2. The procyclicality is asymmetric

Splitting the macro axis at neutral, ratings fall much faster going into recession than they rise going into a boom. Pooled, the downside half-slope (neutral to severe recession) is 0.67 (95% CI [0.63, 0.72]) versus an upside half-slope (boom to neutral) of 0.21 (95% CI [0.18, 0.24])—a 3.2× asymmetry that holds for every model (Figure 2). This is the financially dangerous margin: downturn downgrades dominate, exactly the pattern that transmits rating-driven capital

procyclicality, and it mirrors the stricter-in-downturns behaviour documented for human agencies.

4.3. “Through-the-cycle” instructions anchor rather than de-cyclify

Prompt framing moves the coefficient monotonically: point-in-time 0.516, terse 0.429, “be stable” 0.329, and through-the-cycle 0.160 (95% CI [0.14, 0.18]); the terse-minus-TTC contrast is 0.269 notches per step. Taken alone, this looks like a successful mitigation. It is not. A genuine through-the-cycle rater holds the rating steady while its point-in-time PD still moves with the cycle; anchoring flattens both. Under the through-the-cycle instruction, the PD slope collapses to 0.22–0.59 of its terse value across models—the instruction suppresses the very quantity that should remain cycle-sensitive. Revealingly, the vaguer “be stable” instruction, which is not semantically about the cycle, retains far more PD movement while still calming the rating. The explicit through-the-cycle instruction therefore stabilises the wrong object, and a residual downgrade survives everywhere—up to 1.62 notches across the cycle for GPT-4o-mini.

5. Conclusion

Using a design in which fundamentals are fixed by construction, we provide the first controlled measurement of rating procyclicality in large language models. Frontier LLMs downgrade identical firms as the macroeconomy is described as worsening, by economically meaningful and statistically robust margins, and they do so asymmetrically—punishing simulated downturns far more than they reward simulated expansions. The magnitude is a first-order function of model identity, so in a deployed pipeline the choice of model is itself a prudential decision. The natural mitigation—instructing the model to rate through the cycle—fails instructively: rather than holding the rating steady while letting the default probability track conditions, it suppresses both, and it leaves a measurable residual. These results should be read against their scope—a synthetic firm battery built for identification rather than external realism, five mini/flagship models from two families, and prompt-level rather than fine-tuning interventions—and they motivate benchmarking LLM procyclicality against agency and IRB baselines, testing fine-tuning and retrieval remedies that target the point-in-time object directly, and, in the interim, treating model selection and prompt framing as governed parameters wherever LLMs touch credit and capital decisions.

Declaration of generative AI use

During the preparation of this work the authors used AI-assisted coding and drafting tools to help implement the data pipeline and draft the manuscript. After using these tools the authors reviewed and edited the content and take full responsibility for the content of the published article. The large language models studied are the object of the research, not tools of manuscript preparation.

Declaration of competing interests

The authors declare no competing interests. [Confirm at submission.]

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRedit author statement

Samir Chincholikar: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. Robin Chawla: Conceptualization, Validation, Writing – review and editing, Supervision. [Confirm/adjust at submission.]

Data availability

All data and code are openly available in a Zenodo deposit (DOI to be inserted); reproduction is offline, deterministic and free (see the artifact README and reproduce.sh).

References

- Altman, E.I., 1968. Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *Journal of Finance* 23 (4), 589–609.
- Amato, J.D., Furfine, C.H., 2004. Are credit ratings procyclical? *Journal of Banking & Finance* 28 (11), 2641–2677.
- Bank for International Settlements, 2024. The intelligent financial system: how AI is transforming finance. BIS Working Paper No. 1194. [Verify at submission.]
- Binz, M., Schulz, E., 2023. Using cognitive psychology to understand GPT-3. *Proceedings of the National Academy of Sciences* 120 (6), e2218523120.
- Bommasani, R., Creel, K.A., Bansal, A., Guha, S., Liang, P., 2022. Picking on the same person: does algorithmic monoculture lead to outcome homogenization? *NeurIPS* 35.
- Cantor, R., Packer, F., 1996. Determinants and impact of sovereign credit ratings. *Federal Reserve Bank of New York Economic Policy Review* 2 (2), 37–53.
- Chincholikar, S., Chawla, R., 2026. Do LLMs rate through the cycle? — reproducibility artifact [data set + software]. Zenodo. <https://doi.org/<ZENODO-DOI>>
- Gordy, M.B., Howells, B., 2006. Procyclicality in Basel II: can we treat the disease without killing the patient? *Journal of Financial Intermediation* 15 (3), 395–417.
- Graham, J.R., Harvey, C.R., Jha, M., 2026. CFOs meet LLMs. Working paper, SSRN. [Insert SSRN id/DOI at submission.]
- Horton, J.J., 2023. Large language models as simulated economic agents. NBER Working Paper 31122.
- Kadavath, S., Conerly, T., Askill, A., et al., 2022. Language models (mostly) know what they know. *arXiv:2207.05221*.
- Kashyap, A.K., Stein, J.C., 2004. Cyclical implications of the Basel II capital standards. *Economic Perspectives (Federal Reserve Bank of Chicago)* 28 (1), 18–31.
- Kim, A., Muhn, M., Nikolaev, V.V., 2024. Financial statement analysis with large language models. *arXiv:2407.17866*.
- Löffler, G., 2004. An anatomy of rating through the cycle. *Journal of Banking & Finance* 28 (3), 695–720.
- Löffler, G., 2013. Can rating agencies look through the cycle? *Review of Quantitative Finance and Accounting* 40 (4), 623–646.
- Lopez-Lira, A., Tang, Y., 2023. Can ChatGPT forecast stock price movements? Return predictability and large language models. *arXiv:2304.07619*.
- Tversky, A., Kahneman, D., 1974. Judgment under uncertainty: heuristics and biases. *Science* 185 (4157), 1124–1131.

Tversky, A., Kahneman, D., 1981. The framing of decisions and the psychology of choice. *Science* 211 (4481), 453–458.

Table 1

Per-model rating procyclicality on byte-identical fundamentals. β = notches of downgrade per +1 macro-severity step (terse framing), with firm-clustered bootstrap 95% CIs; all exclude zero. Swing = boom $\rightarrow$ severe-recession notches. Down/Up = downside (into recession) vs upside (into boom) half-slopes. Residual = β after the through-the-cycle instruction; PD-ret. = PD-slope retention under that instruction.

Model	β (terse)	95% CI	Swing	Down	Up	D:U	Residual (TTC)	PD-ret.
Gemini 3.5 Flash	0.230	[0.19, 0.27]	0.96	0.38	0.10	4.0	0.059	0.22
Gemini 2.5 Flash-Lite	0.285	[0.25, 0.32]	1.15	0.45	0.12	3.6	0.056	0.34
GPT-4.1-mini	0.471	[0.41, 0.53]	1.94	0.77	0.20	3.9	0.096	0.24
GPT-4o	0.387	[0.34, 0.43]	1.55	0.59	0.18	3.3	0.187	0.52
GPT-4o-mini	0.775	[0.69, 0.86]	3.18	1.15	0.44	2.6	0.402	0.59

Table 2

Pooled procyclicality by prompt framing (five models), notches per severity step with firm-clustered bootstrap 95% CIs.

Prompt framing	Pooled β	95% CI
Point-in-time	0.516	[0.49, 0.54]
Terse	0.429	[0.41, 0.45]
“Be stable”	0.329	[0.30, 0.35]
Through-the-cycle	0.160	[0.14, 0.18]

Figure 1

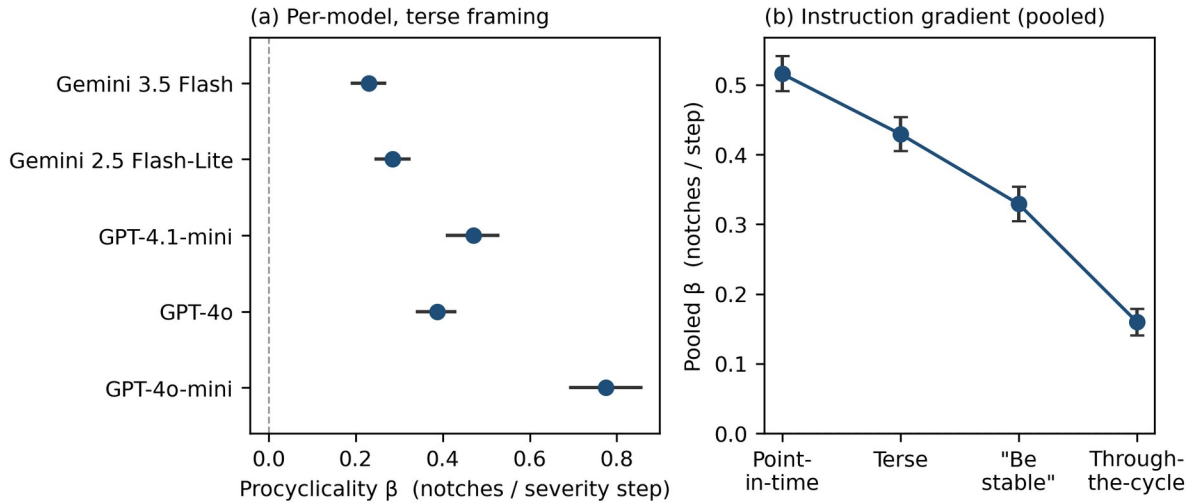


Figure 1. (a) Per-model procyclicality β under the terse framing with 95% CIs; the dashed line marks the through-the-cycle null ($\beta = 0$). (b) Pooled β across the four prompt framings (point-in-time $\rightarrow$ terse $\rightarrow$ "be stable" $\rightarrow$ through-the-cycle) with 95% CIs.

Figure 2

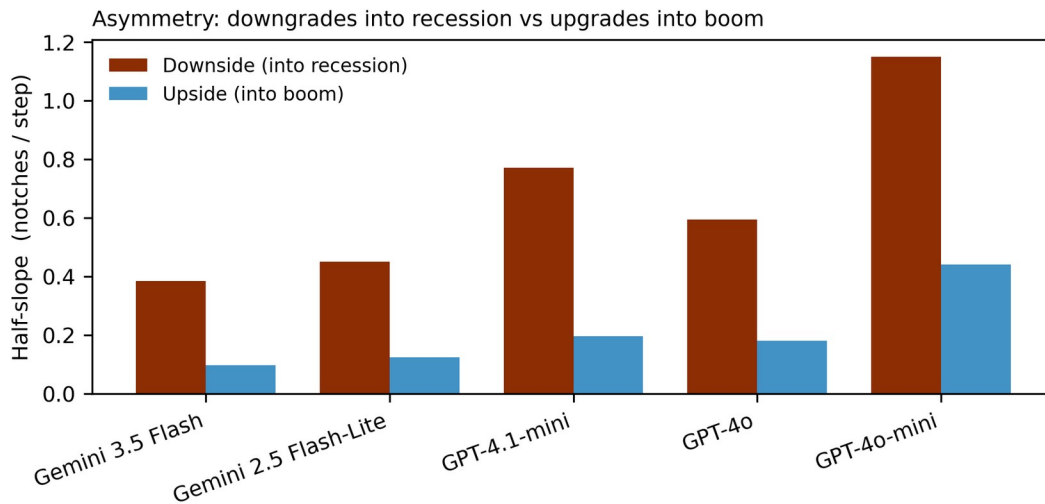


Figure 2. Asymmetry: per-model downside half-slope (neutral $\rightarrow$ severe recession) versus upside half-slope (boom $\rightarrow$ neutral), terse framing. Every model downgrades far faster into recessions than it upgrades into booms.